DNA WEEK 2026 EVENT REPORT

Raising Awareness and Promoting Biotechnology in Rwanda



Organized by:
Department of Biology, School of Science, College of Science and Technology (UR-CST)

In Partnership with:
Bio Cap Rwanda • Alliance for Science Rwanda • AATF • OFAB Rwanda Chapter • RAB

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DNA Week 2026 brought together students, researchers, lecturers, institutions, and partners to explore the growing importance of DNA and biotechnology in shaping agriculture, medicine, research, environmental management, and innovation in Rwanda. The seminar created a platform for discussion, learning, and reflection on how biotechnology can contribute to solving real-world challenges while also strengthening scientific awareness among students and the wider community.

The event featured presentations from experts representing Rwanda Biomedical Centre (RBC), Rwanda Agriculture and Animal Resources Development Board (RAB), REMA, private industry, and biotechnology researchers. Through these sessions, participants gained deeper insight into the practical applications of biotechnology and the importance of responsible scientific advancement.



Fig: Student at University of Rwanda College of Science and Technology.

1. Introduction

DNA Week is a global celebration dedicated to raising awareness about the importance of DNA, biotechnology, and scientific innovation in society. The celebration also commemorates two major milestones in science: the discovery of the DNA double helix structure in 1953 and the completion of the Human Genome Project in 2003.

In Rwanda, the University of Rwanda – College of Science and Technology (UR-CST), through the Department of Biology, organized the third DNA Week celebration under the theme of promoting biotechnology awareness and strengthening public understanding of DNA-based technologies.

The event brought together students, researchers, educators, government institutions, private sector representatives, and development partners to discuss the growing role of biotechnology in agriculture, medicine, environmental management, research, and innovation.

The seminar also created an opportunity for students to interact directly with experts from different institutions and better understand how biotechnology is being applied to solve real challenges in Rwanda.

2. Objectives

The DNA Week 2026 celebration aimed to raise awareness about modern biotechnology and the important role DNA plays in the lives of individuals and society. The event also sought to create a platform where students and participants could interact directly with scientists and professionals working in different biotechnology-related sectors.

Through presentations and discussions, participants were introduced to recent developments in medical biotechnology, agricultural biotechnology, environmental biotechnology, industrial biotechnology, and biosafety systems. The event also encouraged students to think beyond classroom theory and understand how science and innovation can be applied to solve challenges facing communities and the country at large.

3. Event Overview

The event was held at the University of Rwanda – College of Science and Technology and attracted a large number of students and invited guests. The seminar followed a structured agenda that included opening remarks, keynote presentations, expert discussions, and a panel discussion.

The atmosphere throughout the event was interactive and engaging. Students actively participated by asking questions, sharing opinions, and interacting with guest speakers from different sectors.

The seminar focused on how DNA-based technologies are increasingly becoming important tools for addressing challenges in agriculture, healthcare, environmental protection, diagnostics, research, and national development. Throughout the event, students were exposed to practical

examples showing how biotechnology is already being applied in Rwanda to improve crop production, strengthen disease diagnosis, support research systems, and promote innovation.

The sessions also encouraged students to think critically about the future of biotechnology and the role young scientists can play in advancing scientific development in Rwanda.

3.1. Opening Session and Welcome Remarks

The opening session officially welcomed participants, guest speakers, students, and partners attending the DNA Week celebration.

In the welcome remarks delivered on behalf of the Principal of the College of Science and Technology, DNA was described as one of the most remarkable discoveries in science and the foundation of life itself. The remarks emphasized that DNA is not only important in understanding biology but also plays a major role in innovation, medicine, agriculture, conservation, and research.

Participants were encouraged to actively engage throughout the session by asking questions, sharing ideas, and staying curious. Students were reminded that science becomes meaningful when knowledge is connected to solving real-life problems.

The remarks also highlighted the importance of interdisciplinary collaboration, especially the connection between mathematics, genetics, bioinformatics, and biotechnology research. Examples were shared on how mathematical concepts such as statistical modelling, population genetics, graph theory, and bioinformatics contribute to DNA analysis and modern scientific discoveries.

The opening session further stressed that biotechnology should not only remain in laboratories or classrooms, but should contribute directly to improving society and addressing national challenges.



Fig: Prof BANZI Wellars

3.1.Presentation from Medical Biotechnology, Rwanda Biomedical Centre (RBC)

They explored the history of DNA discovery and the contribution of biotechnology in medicine and public health. The speaker explained that understanding DNA was a gradual scientific journey shaped by the contribution of several scientists over time. Participants learned how discoveries around DNA structure and genetic composition opened the way for major developments in medicine and biotechnology.

The presentation highlighted how biotechnology is currently used in healthcare systems through molecular diagnostics, disease detection, recombinant DNA technology, and personalized medicine. One example discussed was the production of insulin using microorganisms, showing how biotechnology can support treatment for diseases such as diabetes.

Participants were also introduced to modern laboratory techniques including DNA extraction, amplification, PCR technologies, and gene-related research methods. The speaker emphasized the growing importance of molecular diagnostics in disease detection and outbreak response, particularly during health emergencies.

The session further demonstrated that biotechnology is no longer limited to laboratories or research institutions, but has become an important part of healthcare systems and scientific innovation. Students were encouraged to view biotechnology as a field capable of improving medical services, strengthening research, and supporting future scientific discoveries.



Fig: Guest Speaker from RBC

3.3. Presentation On BioCap Initiative for Agriculture Research and Extension Alignment

Dr. Bellancile Uzayisenga, Plant and Microbial Biotechnology Program Coordinator under the BioCAP Initiative, delivered a presentation on the role of biotechnology in transforming agriculture in Rwanda.

The presentation introduced the BioCAP project, a five-year initiative supported through international collaboration to strengthen biotechnology research and innovation in Rwanda.

A major component of the project is the establishment of a Biotechnology Centre of Excellence at RAB Rubona. The center aims to strengthen laboratory capacity, support research, and train scientists in biotechnology applications.

The presentation focused on three major crops:

* Potato

* Cassava

* Banana

Participants learned how biotechnology is being used to improve crop resistance against diseases and pests.

Examples presented included:

* Potato varieties resistant to late blight disease

* Cassava resistant to cassava brown streak virus

* Banana gene editing initiatives targeting disease resistance

Dr. Bellancile explained that the project also focuses heavily on capacity building. Scientists and technicians are receiving specialized training from international research institutions to strengthen Rwanda’s biotechnology expertise.



Fig: Dr. Bellancile Uzayisenga

3.4. Presentation On Biosafety Framework in Rwanda

The presentation from REMA focused on biosafety and the importance of regulations in biotechnology development. The speaker explained that while biotechnology offers important opportunities for innovation and problem-solving, there is also a need to ensure that these technologies are developed and applied responsibly.

Participants were introduced to biosafety principles, laboratory safety procedures, risk management systems, and regulations governing genetically modified organisms. The session explained how biosafety frameworks help protect human health, biodiversity, and the environment while supporting scientific progress.

The presentation also highlighted Rwanda's biosafety systems and the country's engagement with international agreements such as the Cartagena Protocol on Biosafety. Students learned that biotechnology regulation is important in ensuring that scientific innovations are carefully assessed before being introduced into communities or ecosystems.

One important issue discussed during the session was public perception and misinformation surrounding GMOs. The speaker encouraged students and scientists to communicate scientific information in simple language that ordinary citizens can understand. Participants were reminded that one of the biggest challenges in biotechnology is not only developing innovations, but also helping society understand them correctly.



Fig: Guest Speaker from REMA

3.5. Presentation from Inter Business Company

The speaker introduced the company's work in supporting laboratories, research institutions, and government systems through DNA technologies and diagnostic platforms.

Participants were guided through the workflow of DNA analysis, including:

* Sample collection
* Sample preparation
* DNA extraction
* Quantification
* Amplification
* Data analysis

The presentation also introduced modern laboratory technologies such as automated nucleic acid extraction systems and PCR platforms used in diagnostics and disease surveillance.

Examples were shared on how these technologies supported Rwanda during disease outbreaks and laboratory operations.

One of the most engaging moments during the session was an interactive quiz where students answered questions related to DNA structure, location, and function. This created a lively learning environment and encouraged active participation.





3.6. Presentation on Engineering Resilient Crops: Biotechnology in Action.

Dr. Léonce DUSENGEMUNGU delivered the presentation focused on how biotechnology is being used to solve real agricultural problems affecting farmers.

He challenged students to become more interested in scientific innovation and biotechnology research, encouraging them to follow scientific developments with the same enthusiasm often given to entertainment and social media trends.

The presentation explained practical examples of biotechnology applications in Rwanda, including:

* Potatoes resistant to late blight disease
* Cassava resistant to cassava brown streak virus
* Maize resistant to stem borers

Participants learned how scientists identify useful genes from organisms and introduce them into crops to improve disease resistance.

He explained complex biotechnology concepts in a simplified way, helping students understand:

* DNA extraction
* Gene identification
* Genetic transformation
* Bioinformatics applications
* Crop improvement techniques



Fig: Dr. Léonce DUSENGEMUNGU

3.7. Panel Discussion and Collaboration Opportunities

The panel discussion provided an opportunity for speakers and participants to discuss collaboration, research opportunities, student engagement, and the future of biotechnology in Rwanda.

Panelists emphasized the importance of strengthening partnerships between universities, research institutions, industry, and government agencies.

Several opportunities for collaboration were discussed, including:

* Student internships
* Joint research projects
* Laboratory access
* Technical training
* Capacity building programs
* Biotechnology awareness activities

Representatives from different institutions encouraged students to actively participate in biotechnology initiatives and seek practical experience beyond classroom learning.

The discussion also emphasized the importance of science communication. Students and young scientists were encouraged to simplify biotechnology concepts and help communities understand the benefits and safety measures associated with biotechnology.

Institutions highlighted ongoing support for students through:

* Training workshops
* Short-term research programs

* Internships at RAB and partner institutions
* Access to research equipment and laboratory facilities

The panel concluded with a strong message that Rwanda's biotechnology future depends not only on technology and infrastructure but also on developing skilled young scientists who can contribute to innovation and public understanding.



Fig: Panel Discussion

4. Conclusion

The DNA Week 2026 celebration at the University of Rwanda, College of Science and Technology successfully created awareness about the importance of DNA and biotechnology in modern society.

Through presentations, discussions, and interactive sessions, participants gained a deeper understanding of how biotechnology is being used in medicine, agriculture, environmental management, diagnostics, and research.

The seminar also demonstrated Rwanda's growing commitment to scientific innovation, research collaboration, and biotechnology development.

Most importantly, the event encouraged students to become active contributors in science, research, and innovation. It reminded participants that biotechnology is not only about scientific discovery, but also about solving real challenges and improving lives.

The event ended with appreciation to all speakers, organizers, students, and partners who contributed to making the third DNA Week celebration successful.